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*Ευχαριστούμε τις εκδόσεις Παπαζήση
για την έκδοση των πρακτικών της διημερίδας*

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COVID-19 PANDEMIC CRISIS: ACCELERATOR OR DECELERATOR OF THE NEW GLOBALIZATION?

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ABSTRACT

The current pandemic crisis of COVID-19 seems to be transforming the global socio-economic system in all dimensions and multiple directions. In this article, we argue that COVID-19 constitutes a “crisis in crisis” because it occurred at a time of an unfolding global transition. After a brief overview of the post-war phases of the evolution of global capitalism, we conclude that the impact of COVID-19 is catalytic, acting as an accelerator of developments. In this context, organizational adaptation and readaptation at all levels are imperative.

Keywords: *COVID-19 crisis, global crisis, postwar development, fourth industrial revolution*

1. Introduction

Towards the end of 2019, a new form of pneumonia appeared in Wuhan, China, now called “COVID-19,” which the World Health Organization has declared an epidemic (Poon & Peiris, 2020). This global health crisis, which has transformed into a global structural socio-economic change, has brought about a crisis like no other (IMF, 2020), the recessionary depth of which might exceed the 2008-2009 economic crisis (Azevêdo, 2020; World Bank, 2020). As it seems, the current pandemic crisis is changing the global socio-economic system profoundly. In this context, this article aims to examine whether COVID-19 accelerates the advent of new developments.

2. A brief history of the post-war development of capitalism

According to Vlado (2019b), we can understand global capitalism’s history by analyzing specific evolutionary phases. In particular, the author suggests distinguishing three co-evolving levels in studying today’s global reality as this develops after WWI by considering how international regimes are structured and how the dynamics of development and crisis unfold. Additionally, this analysis investigates the dominant forms of innovation by using, to a great extent, the Schumpeterian and neo-Schumpeterian contributions to innovation (Chatzinikolaou & Vlado, 2019; Schumpeter, 1934, p. 117).

- According to Krasner (1983), an international regime is a set of implicit or explicit principles, norms, rules, and decision-making procedures around which actors’ expectations converge. In other words, it concerns the forms of cooperation and understanding in the international and global governing institutions and relations that provide global polity with a stable agreement and consensus framework over a prolonged period.
- The development and crisis model results from the “accumulation regime” of quantities and qualities built by national socio-economic systems, accompanied by corresponding regulation forms. The latter is an approach that has been studied and structured under the prism of Fordism and later analyses (Boyer & Durand, 1993; Watson, 2019).

- At the innovation level, the corresponding evolutionary theoretical framework analyzes—in an always-specific space-time context—the dominant forms or “archetypes” of innovation. These archetypes make up the “physiology” or “DNA” of the organization, which allows innovation to sprung up, providing thus the socio-economic organization with survival potential in the ever-mutating conditions of the internal and external environment (Prange & Schlegelmilch, 2010).

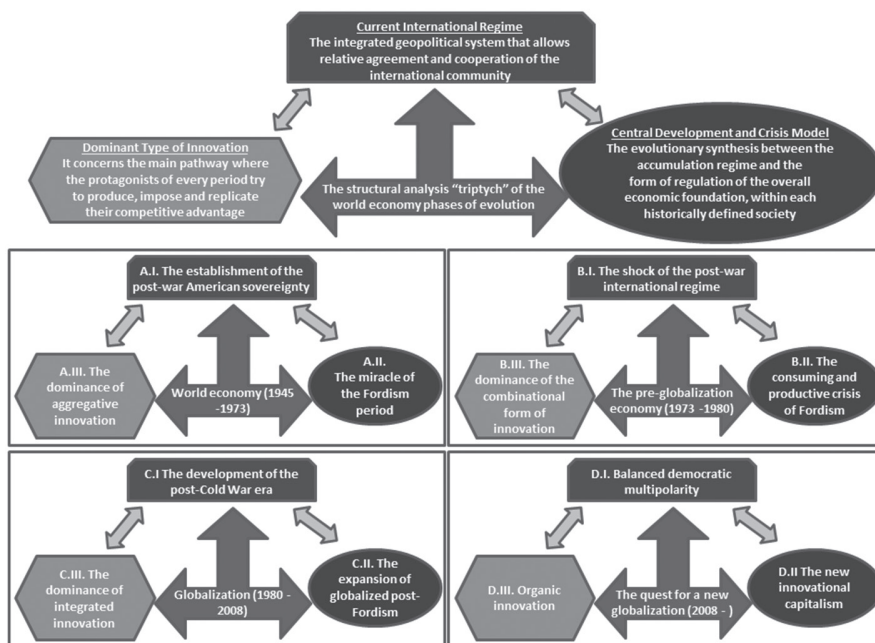


Figure 1: The “structural analysis” of the world economy, according to Vlado and Chatzinikolaou (2020).

In this context, the post-war phases of capitalism have unfolded as follows:

1. Between 1945 and early 1970, the US set up their hegemonic role based on the Bretton Woods agreement (or consensus). At the same time, the dominant growth (and development; Brinkman, 1995) pattern of “Western capitalism” follows the Fordist mass production and consumption (Boyer, 1986; North & Thomas, 1973). Innovation is an issue of organizational ag-

gregation of different elements, where high specialization within larger and nation-centered enterprises plays a decisive role by having a clear and robust Tayloristic imprint (Coriat, 1994).

2. A milestone in the outgoing previously sufficient stabilizing phase is the fixed exchange rate system abolishment. One of the causes of this transforming event was the Middle East oil-rich nations' embargo in retaliation for US support for Israel. This embargo directly affected energy prices, leading to a parallel increase in unemployment and price inflation (stagflation). In this context, the Bretton Woods consensus that dominated international financial and political relations is no longer efficient (Berry et al., 2016). The rapid movement of international capital and US firms' multi-nationalization marks this pre-globalization phase between 1970-1980 (Michalet, 1976). Fordism is entering an identity crisis as the standard of living in the Western world has already increased; consumers are "feverishly" interested in improving their overall socio-economic choices and presence (Lipietz, 1987). In this context, most firms innovate by combining older internal elements, which is increasingly ineffective as industries begin to globalize (Mann, 2011).
3. During the later phase of globalized capitalist development, between 1980-2010, the US hegemony and the Western way of life have a universal character due to the collapse of the rival power of "real socialism" (Gaddis, 1992). A variety of capitalisms appear (Hall & Soskice, 2001), and production is now a case of composing elements not defined in their national contour solely (Paganetto & Scandizzo, 2017; Thursfield, 2017). Innovation is now an issue of cross-fertilization of different production units, industries and socio-economic spaces (Nelson, 1993; Porter, 1998).
4. After the US mortgage crisis outbreak, capitalism's latest development phase has been unfolding (Greenspan, 2008). As showed by this brief overview, the outgoing phase of globalization has also matured structurally, gradually giving way to a new regime, which does not yet have a complete "functional" outline (Laudicina & Peterson, 2016; Vlado et al., 2018). One of the characteristics of the "new globalization" that is emerging is that the international regime is going to be increasingly multipolar, with different "niches" of hegemony created in different corners of the world, such as the BRICS (Rewizorski, 2015). This global capitalism of

our days is a regime of “endless transition” to the knowledge-based society (Etzkowitz & Leydesdorff, 1998), a process of permanent “creative destruction” (Metcalf, 2006). Thus, innovation means a steady demand for higher efficiency and performance, which cannot occur in the absence of change management mechanisms, perceived now mostly from an organic rather than a mechanistic and static perspective (Vlados, 2019a).

Therefore, it seems that an ongoing global crisis somehow bounds the world—a moment in the evolution of global capitalism that resembles an incessant transition, with no prescribed end. The current pandemic of COVID-19 seems to influence this transition radically, acting as a development catalyst.

3. The socio-economic crisis of COVID-19, its effects and the first results

Loss of investment might reach up to 30-40% compared to 2019 (UNCTAD, 2020), and the fall in world trade will record a respective percentage fall (World Economic Forum, 2020a). In this context, humanity might face famines and the rise of extreme poverty again, despite the previously recorded considerable progress on these fronts (United Nations, 2020). Simultaneously, COVID-19 and lockdowns caused the loss of working hours that correspond to 400 million full-time jobs worldwide (ILO, 2020b). Besides these macro-economic data, we can also distinguish the following about the evolution of the global socio-economic system:

- In terms of coordination and global governance, it seems that our world is experiencing a setback. In the analysis, understanding, and action against the current crisis, relative delays and inadequacies have appeared in the World Health Organization, the United Nations, and the Organization for Economic Co-operation and Development. Therefore, this crisis leads to a future “adaptation crisis” and the restructuring of international institutions (G7, G20) in the new post-COVID-19 reality. Instead of the global pandemic crisis rallying forces of coordinated response, the rise of polarization and the “return of the nation-state” at the forefront are apparent (Gaub & Boswinkel, 2020; Grinin & Korotayev, 2020; Landesmann, 2020; Larionova & Kirton, 2020).

- The fourth industrial revolution seems to be at the epicenter of these developments. According to K. Schwab (2016), who has popularized the term to a significant extent, the fourth industrial revolution is that condition where biological or physical systems become increasingly involved with intangible and online systems (cyber-physical systems). After COVID-19, there is an acceleration in this direction in various industries and manifestations of our socio-economic symbiosis and reality. As the International Labor Organization notices (2020a), one in four professions in developed countries is compatible with working from home, while in the developing countries, only one in six is compatible under today's circumstances. The shift is clear: the logic of social distancing and the digitization of work are significant forces among economies and societies. This trend, the intensity of which increased with COVID-19, is also evident because most countries worldwide have taken at least one policy measure after COVID-19 to promote distance learning (UNESCO et al., 2020). Simultaneously, in April 2020, e-commerce increased by 30% compared to April 2019 (OECD, 2020).
- At the organizational adaptation level, most industries undergo digital transformation. According to a sample survey by the World Economic Forum (2020b), most entrepreneurs declare that they have been promoting after COVID-19 digitization and remote working. According to Parker (2020), remote working managers face increasing human resources management problems, a situation that leads to ineffectiveness.

4. Final remarks

This article aimed to investigate, from an introductory perspective, the development of the global socio-economic system after COVID-19. According to the French president, E. Macron (2020), the current pandemic crisis accelerates developments. As he points out, we face an unprecedented situation in which we gradually realize that globalization as we have known it will cease to exist, even though millions of people managed to escape extreme poverty by beating together the forces of authoritarianism. In this context, we arrive at the following conclusions:

- The current global socio-economic crisis created by the COVID-19 pandemic seems to be drastically accelerating the transition towards the new globalization, which has been “hatching” in recent years within the global socio-economic system.
- It acts as a catalyst for the emergence of new challenges for all socio-economic organizations (private and public), cross-industrially and cross-spatially.
- It creates the basis for repositioning international geopolitical affairs nowadays and urgently poses the challenge of renewing the regulatory mechanisms at the national, subnational, and supranational levels.

From a “Gramscian” perspective (Gramsci, 1971), we could argue that the present crisis lies precisely at the point where an earlier order begins to fade, and the new one is struggling to appear (Passet, 1979). Given that an uncertain future is clear, we would suggest organizational adaptation as the primary solution for exiting the multilevel crisis. According to the Darwinian interpretation, the most adaptive species survive, not the strongest ones (Hodgson, 2013). Nowadays, this adaptation consists of improving the organization’s strategy (figure out the present and desired position of the organization in the future); the organization’s technology (how the organization creates and diffuses knowledge); and organization’s management (to learn how to manage even more effectively the scarce resources) (Vlados & Chatzinikolaou, 2019).

References

- Azevêdo, D. G. (2020). Trade set to plunge as Covid-19 pandemic upends global economy. *WTO Trade Forecast Press Conference*, 8. https://www.wto.org/english/news_e/pres20_e/pr855_e.htm [Accessed December 1st, 2020]
- Berry, A., Culpeper, R., & Stewart, F. (2016). *Global development: Fifty years after Bretton Woods*. London, UK: Palgrave Macmillan.
- Boyer, R. (1986). *La théorie de la régulation: Une analyse critique*. Paris: La Découverte.
- Boyer, R., & Durand, J.-P. (1993). *L’après-Fordisme*. Paris: Syros.

- Brinkman, R. (1995). Economic growth versus economic development: Toward a conceptual clarification. *Journal of Economic Issues*, 29(4), 1171–1188. <https://doi.org/10.1080/00213624.1995.11505746>
- Chatzinikolaou, D., & Vlado, Ch. (2019). Schumpeter, neo-Schumpeterianism, and Stra.Tech.Man evolution of the firm. *Issues in Economics and Business (International Economics and Business)*, 5(2), 80–102. <https://doi.org/10.5296/ieb.v5i2.16097>
- Coriat, B. (1994). *L'Atelier et le chronomètre: Essai sur le taylorisme, le fordisme et la production de masse*. Paris: C. Bourgois.
- Etzkowitz, H., & Leydesdorff, L. (1998). The endless transition: A “triple helix” of university-industry-government relations: Introduction. *Minerva*, 36(3), 203–208. <https://doi.org/10.1023/A:1004348123030>
- Gaddis, J. L. (1992). International relations theory and the end of the Cold War. *International Security*, 17(3), 5–58. <https://doi.org/10.2307/2539129>
- Gaub, F., & Boswinkel, L. (2020). *The geopolitical implications of the COVID-19 pandemic* [Policy Department for External Relations: Directorate General for External Policies of the Union]. Brussels: European Parliament.
- Gramsci, A. (1971). *Selections from the Prison Notebooks* (11th printing: 1992). New York, US: International Publishers.
- Greenspan, A. (2008). *The age of turbulence: Adventures in a new world*. London, UK: Penguin.
- Grinin, L., & Korotayev, A. (2020). *Covid-19 Pandemic, Geopolitics, and Recession* (Moscow, Russia: International Center for Education and Social and Humanitarian Studies Working paper #4). Lomonosov Moscow State University, Russia.
- Hall, P. A., & Soskice, D. W. (Eds.). (2001). *Varieties of capitalism: The institutional foundations of comparative advantage*. Oxford, UK; New York, US: Oxford University Press.
- Hodgson, G. (2013). Understanding organizational evolution: Toward a research agenda using generalized darwinism. *Organization Studies*, 34(7), 973–992. <https://doi.org/10.1177/0170840613485855>
- ILO. (2020a). *Working from Home: Estimating the worldwide potential* [Policy Briefs]. Switzerland: International Labour Organization.
- ILO. (2020b). *ILO Monitor: COVID-19 and the world of work. Fifth edition*. Switzerland: International Labour Organization
- IMF. (2020). *World Economic Outlook, April 2020: The Great Lockdown* (World Economic Outlook Reports).
- Krasner, S. D. (1983). *International regimes*. New York, US: Cornell University Press.
- Landesmann, M. A. (2020). Covid-19 crisis: Centrifugal vs. centripetal forces in the EU—a political-economic analysis. *Journal of Industrial and Business Economics*, 47(3), 439–453. <https://doi.org/10.1007/s40812-020-00171-w>

- Larionova, M., & Kirton, J. (2020). Global governance after the COVID-19 crisis. *International Organisations Research Journal*, 15(2), 7–23. <https://doi.org/10.17323/1996-7845-2020-02-01>
- Laudicina, P. A., & Peterson, E. R. (2016). *From globalization to islandization* (Global Business Policy Council (GBPC) /Research Report). AT Kearney.
- Lipietz, A. (1987). *Mirages and miracles: The crisis in global Fordism*. London, UK: Verso.
- Macron, E. (2020, April 14). Transcript: We are at a moment of truth. *Financial Times*. <https://www.ft.com/content/317b4f61-672e-4c4b-b816-71e0ff63cab2> [Accessed December 1st, 2020]
- Mann, S. (Ed.). (2011). *Sectors matter! Exploring mesoeconomics*. Heidelberg: Springer. <https://doi.org/10.1007/978-3-642-18126-9>
- Metcalfe, J. S. (2006). *Evolutionary economics and creative destruction*. Oxon: Routledge.
- Michalet, C. A. (1976). *Le capitalisme mondial*. Paris: Presses universitaires de France.
- Nelson, R. (1993). *National innovation systems. A comparative analysis*. New York, US; Oxford, UK.
- North, D. C., & Thomas, R. P. (1973). *The rise of the Western world: A new economic history*. Cambridge, UK: Cambridge University Press. <https://doi.org/10.1017/CBO9780511819438>
- OECD. (2020). *E-commerce in the time of COVID-19* (Tackling Coronavirus (COVID-19): Contributing to a Global Effort). Secretary-General of the OECD. <http://www.oecd.org/coronavirus/policy-responses/e-commerce-in-the-time-of-covid-19-3a2b78e8/> [Accessed December 1st, 2020]
- Paganetto, L., & Scandizzo, P. L. (2017). Post-fordism, new economy and the case of the Italian “Mezzogiomo.” In *Knowledge Economy, Information Technologies and Growth* (pp. 501–519). Routledge; <https://doi.org/10.4324/9781351154567-21>
- Parker, S. K., Knight, C., & Keller, A. (2020). Remote managers are having trust issues. *Harvard Business Review*. <https://hbr.org/2020/07/remote-managers-are-having-trust-issues> [Accessed December 1st, 2020]
- Passet, R. (1979). *L' économi que et le vivant*. Paris: Payot.
- Poon, L. L. M., & Peiris, M. (2020). Emergence of a novel human coronavirus threatening human health. *Nature Medicine*, 26(3), 317–319. <https://doi.org/10.1038/s41591-020-0796-5>
- Porter, M. (1998). Location, clusters, and the “new” microeconomics of competition. *Business Economics*, 33(1), 7–13.
- Prange, C., & Schlegelmilch, B. B. (2010). Heading for the next innovation archetype? *Journal of Business Strategy*, 31(1), 46–55. <https://doi.org/10.1108/02756661011012778>

- Rewizorski, M. (2015). *The European Union and the BRICS: Complex relations in the era of global governance*. Cham, Switzerland: Springer. <https://doi.org/10.1007/978-3-319-19099-0>
- Schumpeter, J. (1934). *The theory of economic development: An inquiry into profits, capital, credit, interest, and the business cycle* (R. Opie, Trans.; Reprint Edition: New Brunswick, N.J, Transaction Books, 1983, Social science classics series). Cambridge, MA: Harvard University Press.
- Schwab, K. (2016). *The fourth industrial revolution*. New York: Crown Business.
- Thursfield, D. (2017). *Post-fordism and skill: Theories and perceptions*. London, UK: Routledge. <https://doi.org/10.4324/9781315182711>
- UNCTAD. (2020). *Impact of the COVID-19 pandemic on global FDI and GVCs* (Special Issue March 2020; Investment Trends Monitor). Geneva, Switzerland: UNCTAD.
- UNESCO, UNICEF, & World Bank. (2020). *What have we learnt? Overview of findings from a survey of ministries of education on national responses to COVID-19*. UNESCO, UNICEF, World Bank. <https://openknowledge.worldbank.org/handle/10986/34700> [Accessed December 1st, 2020]
- United Nations. (2020). *Global Humanitarian Response Plan: COVID-19* (United Nations Coordinated Appeal: April – December 2020). <https://www.unocha.org/sites/unocha/files/Global-Humanitarian-Response-Plan-COVID-19.pdf> [Accessed December 1st, 2020]
- Vlados, Ch. (2019a). Change management and innovation in the “living organization”: The Stra.Tech.Man approach. *Management Dynamics in the Knowledge Economy*, 7(2), 229–256. <https://doi.org/10.25019/MDKE/7.2.06>
- Vlados, Ch. (2019b). The phases of the postwar evolution of capitalism: The transition from the current crisis into a new worldwide developmental trajectory. *Perspectives on Global Development and Technology*, 18(4), 457–488. <https://doi.org/10.1163/15691497-12341528>
- Vlados, Ch., & Chatzinikolaou, D. (2019). Business ecosystems policy in Stra.Tech.Man terms: The case of the Eastern Macedonia and Thrace region. *Journal of Entrepreneurship, Management and Innovation*, 15(3), 163–197. <https://doi.org/10.7341/20191536>
- Vlados, Ch., & Chatzinikolaou, D. (2020). BRICS and global restructuring: Notes for the near future. *Management and Economics Research Journal*, 6(S5), 12934. <https://doi.org/10.18639/MERJ.2020.1075592>
- Vlados, Ch., Deniozos, N., Chatzinikolaou, D., & Demertzis, M. (2018). Perceiving competitiveness under the restructuring process of globalization. *International Journal of Business and Management*, 13(8), 135–153. <https://doi.org/10.5539/ijbm.v13n8p135>

- Watson, D. (2019). Fordism: A review essay. *Labor History*, 60(2), 144–159. <https://doi.org/10.1080/0023656X.2019.1537031>
- World Bank. (2020). *Global economic prospects*. Washington, DC: World Bank. <https://doi.org/10.1596/978-1-4648-1553-9>
- World Economic Forum. (2020a, April 9). *Coronavirus could reduce world trade by up to a third, according to the WTO*. World Economic Forum. <https://www.weforum.org/agenda/2020/04/wto-financial-crisis-coronavirus-covid19-recession-trade-global/>
- World Economic Forum. (2020b). *The future of jobs report*. Switzerland: World Economic Forum.

ΠΡΟΓΝΩΣΗ ΤΗΣ ΑΝΕΡΓΙΑΣ ΣΤΗΝ ΕΥΡΩΖΩΝΗ ΜΕ ΜΕΘΟΔΟΥΣ ΜΗΧΑΝΙΚΗΣ ΜΑΘΗΣΗΣ

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ΠΕΡΙΛΗΨΗ

Η ανεργία έχει άμεση επίδραση στα δημόσια οικονομικά και σοβαρές κοινωνικοπολιτικές επιπτώσεις. Συνεπώς, η ακριβής μέτρηση, ο έγκαιρος έλεγχος και μείωση της ανεργίας είναι από τους βασικούς στόχους των κυβερνήσεων. Η μελέτη αυτή αποσκοπεί στην πρόγνωση της κατεύθυνσης του ποσοστού ανεργίας στην ευρωζώνη. Απ' όσο γνωρίζουμε, καμία άλλη μελέτη δεν επιχείρησε να προγνώσει το ποσοστό ανεργίας στην ευρωζώνη συνολικά. Το σύνολο των δεδομένων περιλαμβάνει το ποσοστό ανεργίας και 36 επεξηγηματικές μεταβλητές από 1998:4 έως 2019:9 σε μηνιαία συχνότητα. Οι μεταβλητές αυτές τροφοδοτούνται σε τρεις μεθοδολογίες Μηχανικής Μάθησης (Machine Learning) τα Decision Trees (DT), τα Random Forests (RF) και τα Support Vector Machines (SVM) και στην οικονομετρική μέθοδο Logistic Regression (Logit). Τα βέλτιστα αποτελέσματα επιτυγχάνονται μ' ένα μοντέλο random forest με ακρίβεια πρόγνωσης 88.5% στο σύνολο των δεδομένων και 85.4% στα δεδομένα out-of-sample.